

Modeling the Downstream Processing of Monoclonal Antibodies Reveals Cost Advantages for Continuous Methods for a Broad Range of Manufacturing Scales

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The biopharmaceutical industry is evolving in response to changing market conditions, including increasing competition and growing pressures to reduce costs. Single-use (SU) technologies and continuous bioprocessing have attracted attention as potential facilitators of cost-optimized manufacturing for monoclonal antibodies. While disposable bioprocessing has been adopted at many scales of manufacturing, continuous bioprocessing has yet to reach the same level of implementation. In this study, the cost of goods of Pall Life Science's integrated, continuous bioprocessing (ICB) platform is modeled, along with that of purification processes in stainless-steel and SU batch formats. All three models include costs associated with downstream processing only. Evaluation of the models across a broad range of clinical and commercial scenarios reveal that the cost savings gained by switching from stainless-steel to SU batch processing are often amplified by continuous operation. The continuous platform exhibits the lowest cost of goods across 78% of all scenarios modeled here, with the SU batch process having the lowest costs in the rest of the cases. The relative savings demonstrated by the continuous process are greatest at the highest feed titers and volumes. These findings indicate that existing and imminent continuous technologies and equipment can become key enablers for more cost effective manufacturing of biopharmaceuticals.

1. Introduction

It is well established that the biopharmaceutical industry is expected to face increased pressure to reduce costs associated with all stages of the

drug development lifecycle. High commercial manufacturing costs put an upper limit on the potential for market expansion, as they contribute to unsustainable prices for domestic and global end-users, as well as the institutions that support them.^[1–5] Blurred ethical lines and incidents of fraud in the wider pharmaceutical industry could also increase pressure for lower prices, as negative public perception, if pushed far enough, can lead to implementation of new laws and regulations.^[6–8]

If biopharmaceuticals face pressure for cost reduction, monoclonal antibodies (mAbs) will likely be a target, as they make up one of the largest and fastest growing biologics markets and are relatively expensive to produce.^[9–11] At the early stages of mAb development, uncertainty surrounding a candidate's clinical success makes high production costs a liability.^[12] This issue is magnified by the fact that multiple drugs often exist in a company's clinical pipeline at a given time. After launch, higher than average manufacturing costs hinder success against competition, which could be heightened due to an increased prevalence of biosimilars.^[3,10,13,14]

Single-use (SU) and continuous technologies will likely play some role in addressing the future need for lower mAb manufacturing costs.^[15] SU technology, both upstream and downstream, has been consistently shown to reduce mAb manufacturing costs at many stages and scales.^[2,12,16–19] Utilization of SU technology in mAb processes, especially at small to medium scales, is also not particularly unfeasible or controversial.^[20–24] Continuous bioprocessing has been shown to reduce costs,^[12,25–27] but more practical challenges remain and adoption is still in its infancy.^[28,29]

While continuous bioprocessing offers the potential for reduced cost and increased consistency in product quality, it also creates a need for new regulations and strategies around quality monitoring and release.^[25,30] A notable level of automation is required to run a process for extended periods of time and reap the full benefits of continuous processing, which means development costs falling on either the vendors or end-users of continuous technology.^[14,25,29,31] Given the challenges and uncertainties associated with the adoption of

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continuous bioprocessing, the biopharmaceutical industry needs an ever-growing body of evidence that the reward will be worth the risk.

Cost modeling studies that focus on continuous mAb purification often compare processes that use different upstream formats to target the same mAb demand.^[12,26] Studies that do compare different downstream modes at the same feed scale either do so in detail for a few cases, or do not consider a continuous downstream in which all unit operations after the bioreactor occur in parallel.^[25,27]

The aim of this study is to identify which mode of downstream bioprocessing is most cost effective over a broad range of manufacturing scales. Three downstream processes were modeled using BioSolve Process for direct cost of goods (CoGs) and net present cost (NPC) comparison. The three processes: stainless steel (SS) batch, SU batch, and an integrated, continuous bioprocessing (ICB) platform were analogous in that they followed what is more or less an established set of downstream unit operations for mAb purification.^[27,32,33] Downstream costs were evaluated for the three processes across an array of scenarios relevant to clinical and commercial manufacturing. Costs were only ever compared at the same upstream volume, titer, and number of batches per year, which allowed for a more direct comparison of downstream cost as a function of feed scale.

2. Experimental Section

2.1. Overarching Approach

BioSolve Process v7 (Biopharm Services Limited) was used to model the CoGs of three downstream mAb processes: SS Batch, SU Batch, and the ICB Platform. Cost evaluation of the three manufacturing strategies was conducted for sets of both clinical and full-scale manufacturing scenarios for production of a single mAb. Each set of scenarios consisted of a full factorial design, in which the factors considered were volume to process, titer, and number of batches per year (Table 1). These sets of scenarios are

here referred to as the clinical and commercial design space, respectively. In this study, the three upstream factors are defined in terms of the bulk feed that each downstream process receives, which could be produced by a variety of specific upstream formats. However, the upstream format is intentionally left undefined in this study to focus on the difference in downstream CoGs.

2.2. Upstream Design Spaces

The clinical design space had a throughput range for a single mAb of 0.1–19 kg year⁻¹ after yield losses. This throughput range was attained via bioreactor volumes of 200 to 2000 L, titers from 1 to 5 g L⁻¹, and 1 to 3 batches per year (Table 1). All consumables that tend to be used more than once had relatively limited reuse in this design space. TFF and chromatography membranes, resin, and prepacked columns were assumed to be disposed of after 1 year, corresponding to a maximum of three batches.

The commercial design space had a throughput range for a single mAb of 17–1600 kg year⁻¹, which was attained via bioreactor volumes of 2000 to 12 000 L, titers from 1 to 9 g L⁻¹, and 15 to 25 batches per year (Table 1). A 5% batch failure rate was also assumed for all commercial scenarios. The bioreactor titer range increases in commercial manufacturing because it was assumed that for a given drug and cell line, more time for development means a potential for increased upstream productivity by the time commercial manufacturing is reached. In this design space, sorbent, and prepacked columns for bind and elute steps could be reused for the shelf-life of 3 years, or if it was reached sooner, the maximum number of cycles (200). Cleaning validation capital was included in all three downstream models to support reuse of chromatography related consumables. For the flow-through chromatography steps, membrane adsorbers and sorbent were also reused for up to 3 years, corresponding to a maximum of 75 cycles at 25 batches/year. TFF membranes were reused for 1 year, corresponding to a reuse for a maximum of 25 batches. It should be noted that in either

Table 1. Factors considered for each process in the clinical and commercial design spaces (full factorial).

Clinical factors (0.1–19 kg year ⁻¹)			
Total feed volume/batch (L)	200	1000	2000
Feed titer (g L ⁻¹)	1	3	5
Batches/year	1	2	3
Bind & elute sorbent reuse	200 cycles or 1 year		
Flow-through membrane/sorbent reuse	1 year (3 cycles maximum)		
TFF membrane reuse	1 year (3 batches maximum)		
Commercial factors (17–1600 kg year ⁻¹)			
Total feed volume/batch (L)	2000	6000	12 000
Feed titer (g L ⁻¹)	1	5	9
Batches/year	15	20	25
Bind & elute sorbent reuse	200 cycles or 3 years		
Flow-through membrane/sorbent reuse	3 years (75 cycles maximum)		
TFF membrane reuse	1 year (25 batches maximum)		

design space, no difference in lifetime utilization was assumed between the three downstream processes for flow-through chromatography media or TFF membranes.

2.3. High-Level Assumptions

The following assumptions were implemented for all three process models:

- For all scenarios and processes in this study, capital estimation for a new facility was assumed.
- Default BioSolve facility support equipment were included in the capital, along with default floor area estimation based on the sizes of all unit operations and supporting equipment.
- A 10 year project lifetime was assumed, along with a 10% future value after depreciation and a 12% cost of capital.
- Only downstream operating and capital costs were included in the CoGs calculation. This is because it is assumed that the upstream format, along with the upstream costs, are the same for all three downstream processes. Thus the upstream costs were omitted to focus on highlighting differences in downstream CoGs.
- The total labor cost of each downstream process was calculated using default rates and an annual headcount that was estimated using the total process-related labor hours.
- When offline clean-in-place (CIP) operations were necessary, default labor and material requirements were assumed.

2.4. Downstream Models and Assumptions

Figure 1 depicts the sequence of unit operations modeled for each downstream process. All processes followed the same general order of unit operations, starting with varying forms of clarification, followed by Protein A capture, low pH viral inactivation, two polishing chromatography steps, virus filtration, and formulation utilizing TFF. This is in line with the generally accepted set of downstream mAb purification operations.^[27,32,33] The SS Batch process utilized exclusively SS unit operations, while the SU Batch and ICB Platform processes utilized disposable technologies wherever possible. The only exceptions were use of a centrifuge by the SU Batch process, as well as the use of SS tanks by both processes, at and above the 2000 L scale (see sections 2.4.2 and 2.4.3). The same target output concentration of 150 g L^{-1} constrained the formulation section of all three processes. The overall mAb yield was assumed to be 62% for all three processes. This was based on yields observed for in-house experiments with a lab-scale, continuous process,^[34] as well as typical values accepted in batch mAb purification.^[35]

The upstream feed was assumed to be moderately challenging to clarify with depth filtration, meaning that 40 m^2 of total depth filter area would be required to filter 1000 L of feed. For the Cadence™ Acoustic Separator in the continuous process, an 80% reduction in cells and cellular debris, corresponding to an 80% reduction in depth filter surface area, was assumed.^[34] For processes using a

disc-stack centrifuge, a 90% reduction in depth filter area was assumed.

2.4.1. SS Batch Model

The SS Batch downstream model (Figure 1A) is a reconfigured and modified version of the default “mAb Typical” that is included in BioSolve. It is a fairly traditional downstream mAb process that utilizes high capital SS equipment and often requires CIP. All unit operations occur staggered in time, with the process steps of each unit unable to begin until the previous ones have finished. In this way, the process was configured to take anywhere from 3 to 9 shifts of 8 hr each, depending on the amount of feed to be processed. Process personnel were set to two for all steps of most unit operations, but three personnel were assigned to chromatography and formulation TFF operations. Product hold vessels were assumed between most unit operations and SS buffer preparation, and fluid storage were used. Custom unit operation models were used for the Protein A (Kaneka Corporation’s KanCapA) and mixed-mode (Pall Corporation’s CMM HyperCel™) bind and elute chromatography steps. Performance assumptions for the sorbents came from in-house breakthrough data. Loading was assumed to be 60% of the volume required to achieve 10% mAb breakthrough, corresponding to approximately 1% breakthrough. These sorbents were used in the continuous platform during the proof-of-concept experiments,^[34] so they were assumed here for all three processes to allow for a balanced comparison of the chromatography operations.

2.4.2. SU Batch Model

The SU Batch downstream model (see Figure 1B) is similar to SS Batch in terms of sequential unit operations and product hold requirements, however this process utilized more disposable components, as well as more custom unit operations and price data. Filtration unit operations throughout the process were assumed to be in Pall’s SU formats instead of traditional formats with housings that require cleaning. Specifically, the clarification section of this process used only depth filtration in Pall’s Stax™ format at feed volumes below 2000 L. Due to the feasibility issues associated with the depth filtration of high volumes, a disc stack centrifuge with supporting filtration was assumed for feed volumes above 2000 L. For feed volumes of 2000 L, depth filtration was assumed in the clinical design space and a disc stack centrifuge was assumed in the commercial design space. The bind and elute chromatography steps utilized the same default skids as SS Batch, but with Repligen’s OPUS prepacked columns instead of columns packed in-house. The same breakthrough data and loading assumptions were used as in the SS Batch bind and elute models. Pall’s Mustang® Q anion exchange membranes also replaced a traditional column for the flow-through polishing step. Because the prepacked columns have a maximum inner diameter of 60 cm, a scaling strategy of increasing the number of batch cycles is employed by the model when a column with an inner diameter greater than 60 cm is required. This means that for the highest feed scales the bind

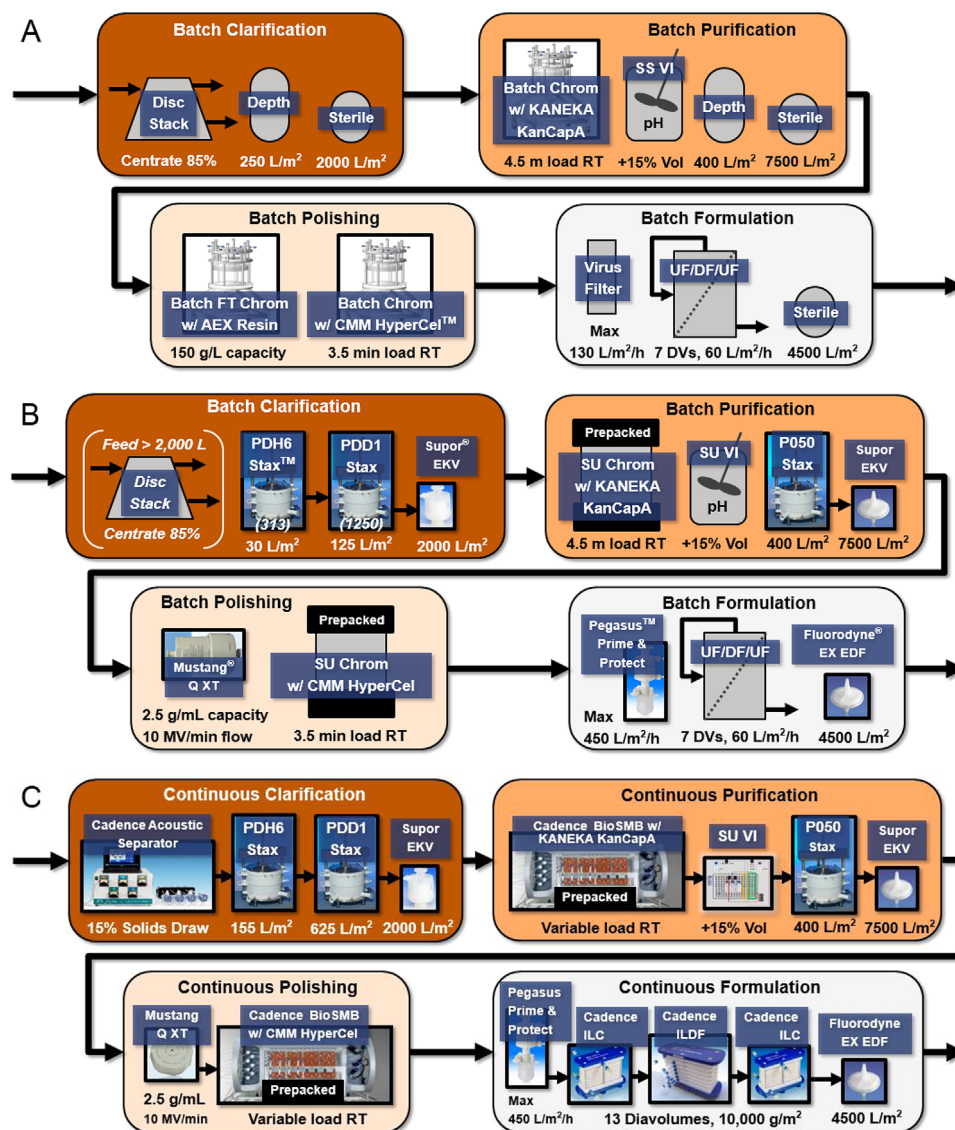


Figure 1. Diagram of the process sequence and key scaling assumptions for the SS Batch (A), SU Batch (B), and ICB Platform (C) models. Images of equipment and consumables are for example only. ILC and ILDF refer to Inline Concentration and Diafiltration, respectively.

and elute chromatography steps had significantly longer processing durations than the analogous units in SS Batch. As a result, the entire SU Batch process took anywhere from 3 to 21 shifts of 8 hr, depending on the feed to be processed. The same process personnel assignment was used here as in the SS Batch model. Default SU buffer preparation and fluid storage vessels were assumed for vessel volumes at or below 2000 L. Above this scale SS vessels were used by the model.

2.4.3. ICB Platform Model

The ICB Platform modeled here (see Figure 1C) utilizes Pall's existing and planned disposable and continuous technologies. All unit operations in this process were assumed to run in parallel. The entire process was assumed to take a maximum of

24–48 hr, with the 48 hr limit required only at the highest commercial volume. Four process personnel were assumed to operate the complete continuous process throughout the entire downstream duration. This is a greater amount of personnel per shift than was assumed for the batch processes, but without further process automation more personnel are required to handle many units running at the same time. The ICB Platform model utilized many of the same disposable filtration products as the SU Batch process, but they were operating continuously over longer durations. Key differences from the SU Batch process that enabled continuous operation were the Cadence Acoustic Separator for primary clarification, the Cadence BioSMB system for capture and polishing chromatography, and the Cadence Inline Concentrator and Diafiltration units for formulation. In addition to enabling continuous operation, the Cadence Acoustic Separator allows for a significant reduction in

clarification depth filtration area. The Cadence BioSMB is a form of multi-column chromatography system that enables continuous operation and reduced sorbent volume via secondary loading columns and parallel column regeneration.^[36–38] The Cadence BioSMB model utilized the same experimental breakthrough data as the batch bind and elute models, but loading was assumed to be 80% of the volume required to reach 10% breakthrough, corresponding to approximately 1% breakthrough. The breakthrough data and loading assumptions were used, along with methods described in previous work,^[39] to determine operating binding capacity for two or three loading columns in series at a given load residence time. Utilizing multiple loading columns allows the BioSMB to reach higher sorbent capacities than single column chromatography, because the primary column reaches an overloaded capacity and the auxiliary column(s) capture the product that has broken through. A set of BioSMB configuration calculations used the resin constraints and feed information to select an operating mode for each scenario that allowed for a significant reduction in total resin volume and prepacked column diameter compared to batch chromatography. It should be noted that many of the scaling parameters used in the continuous unit operation models were based on Pall's in-house, proof-of-concept experiments in which the lab-scale version of the process was used to purify a fed-batch harvest in under 24 hr.^[34] The same disposable strategy for buffer preparation and fluid storage was assumed here as in the SU Batch model, only with significantly reduced requirements for product hold capacity due to the continuous operation of the process.

2.5. Net Present Cost Analysis

For the commercial design space, a comparison was conducted of the NPC of each process at a 5 g L^{-1} titer for the full range of bioreactor volume and batches per year. A 14 year project lifetime was assumed for all three processes, with 10 years of active plant operation. NPC calculation is similar to that of net present value (NPV), where the discounted annual cash flows for the project are used to calculate the present-day value of the project. However, revenue is omitted to only calculate the total cost of the project.^[12] The calculation was achieved here by using BioSolve's NPV tool and setting the mAb sales price to zero. The default tax (36%) and inflation (3%) rates were used. Capital was set to start depreciating on the first year of plant operation and reach a 10% final value. In this study no difference was assumed between the processes for the length of the building period or the capital spending profile before manufacturing begins.

3. Results

The SS Batch, SU Batch, and ICB Platform models detailed in section 2.4 were used to calculate the CoGs for all clinical and commercial scenarios described in section 2.2. The result was a large data set including the CoGs for each process, broken down into contribution by downstream section and cost category, for a wide range of feed scales. Large sections of

this data set can be visualized all at once to identify overarching trends, and individual cases can be examined in detail to see key cost drivers.

3.1. Clinical Cost Results

The clinical design space consisted of a relatively low mAb throughput range, achieved by processing smaller bioreactors at lower titers several times per year with limited reuse of consumables (Table 1). From the clinical data set, contour plots of the annual cost for each downstream process at two batches/year and varying bioreactor titer and volume were estimated (Figure 2). The first thing to note is that annual costs increase with mAb throughput for all three processes, which is to be expected. What is more interesting is the difference in how each downstream process responds to increasing mAb throughput. The SS Batch process has the highest CoGs across the clinical design space, starting out at a relatively high annual cost at 200 L and 1 g L^{-1} and increasing steadily as the process moves toward 2000 L and 5 g L^{-1} (Figure 2A). SU Batch starts out with the lowest CoGs at the lowest throughput, but annual costs increase with throughput more rapidly than SS Batch, particularly in the higher throughput regime (Figure 2B). The ICB Platform starts out at 200 L and 1 g L^{-1} with a low cost relative to SS Batch and is the least sensitive of the three processes to scale via bioreactor titer and volume. It is this decreased sensitivity to increasing mAb throughput that allows the continuous process to emerge as the one with the lowest CoGs in the high throughput half of the clinical design space (Figure 2C).

Every clinical scenario used to generate Figure 2 can be examined with a higher level of resolution to identify which areas of each process contribute most to the cost. Detailed cost results for the processing of two batches/year of 2000 L at 3 g L^{-1} are presented in Figure 3. In this scenario, the ICB Platform exhibit 36% savings over SS Batch and 20% savings over SU Batch, which equates to \$2.7 and \$1.2 M annual savings, respectively. In this case, the majority of the ICB Platform savings come from the clarification and purification sections.

The Cadence Acoustic Separator is responsible for the savings in the clarification section in Figure 3. It allows for a major reduction capital as compared to the disc stack centrifuge used by the SS Batch process. The Cadence Acoustic Separator also provides a reduction in consumables and capital over the clarification used in SU Batch, which relies solely on depth filtration. The floor area and supporting filtration equipment required for depth filtration operations at this scale contribute to the higher capital seen in the SU Batch clarification and the cost of the filters themselves contributes to the consumables.

The Cadence BioSMB system is responsible for the savings seen in the purification section in Figure 3. These savings come entirely from the consumables category, which includes Protein A sorbent and prepacked columns, as well as buffer preparation and storage bags. The Cadence BioSMB system allows for significant increases in sorbent productivity and reductions in buffer utilization over single column chromatography.^[36–38,40] This translates to reduced Protein A volume

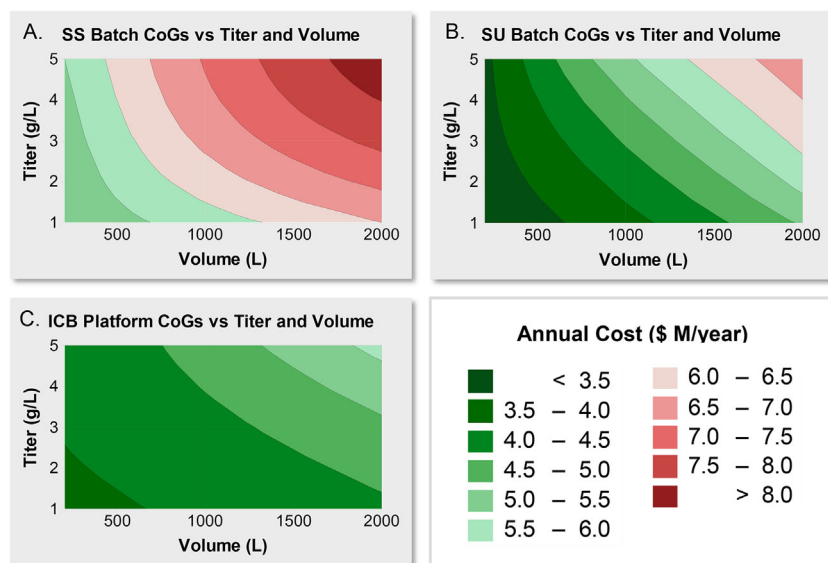


Figure 2. Contour plots of the annual CoGs associated with SS Batch (A), SU Batch (B), and ICB Platform (C) versus bioreactor titer and volume at two batches/year (clinical). No difference in yield was assumed between the three processes, so the annual costs are all effectively normalized by purified mAb throughput.

and buffer storage requirements over the batch processes. The BioSMB can process the same amount of mAb with less sorbent because secondary loading columns, along with parallel column regeneration, enable sorbent to be cycled more rapidly without losing product. The ability of the BioSMB to cycle resin quickly has the highest cost impact in clinical scenarios like the one depicted in Figure 3, where expensive Protein A resin is only used for two batches before disposal. It should also be noted that the columns used here by the BioSMB have a much smaller inner diameter than the column used in the SU Batch process, because a reduced total sorbent volume is distributed across multiple columns. This leads to lower prepacked column costs compared to the SU Batch process because, at this scale, the cost of several smaller diameter columns is less than that of one large column.

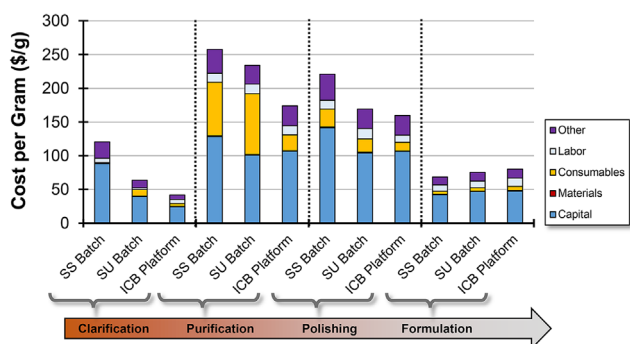


Figure 3. A comparison of the CoGs per gram of each process at a single clinical case: two batches/year of 2000 L at 3 g L^{-1} . The costs are sorted by downstream section, as defined in Figure 1, and standard BioSolve CoGs category.

3.2. Commercial Cost Results

The commercial design space consisted of scenarios with higher mAb throughput than the clinical design space. This was achieved by processing larger bioreactors at higher titers many times per year (Table 1). Consumables were also reused more frequently here than in the clinical scenarios. From the commercial data set, contour plots of the annual cost for each downstream process at 20 batches/year and varying bioreactor titer and volume were estimated (Figure 4). The SS Batch process has the highest CoGs across the commercial design space, as was the case for the clinical scenarios. Annual costs for this process increase sharply in response to increasing mAb throughput, especially in the high titer and high volume regime (Figure 4A). The SU Batch process is the second most costly process and the relative cost reduction compared to SS Batch is reduced at these scales (Figure 4B). It should be noted that for the commercial design space, the SU Batch process becomes more of a hybrid process, still utilizing many disposable components, but also relying on a disc stack centrifuge and SS fluid storage due to feasibility issues. The ICB Platform is also using more SS storage at this scale, but it does not cross the 2000 L threshold as often due to significantly reduced total buffer requirements. The ICB Platform exhibits the lowest CoGs across the entire commercial design space, with the best savings occurring at the highest titers and volumes (Figure 4C). Once again, this is due to the ICB Platform scaling less sensitively with bioreactor titer and volume as compared to the other two processes.

An isolated commercial scenario from Figure 4 can be broken down to identify cost drivers at these larger scales. Detailed cost results for the processing of 20 batches/year of 6000 L at 5 g L^{-1} for the three processes are shown in Figure 5. For this scenario, the ICB Platform exhibits 30% savings over SS Batch and 16% savings over SU Batch, which equates to \$4.7 and \$2.2 M annual

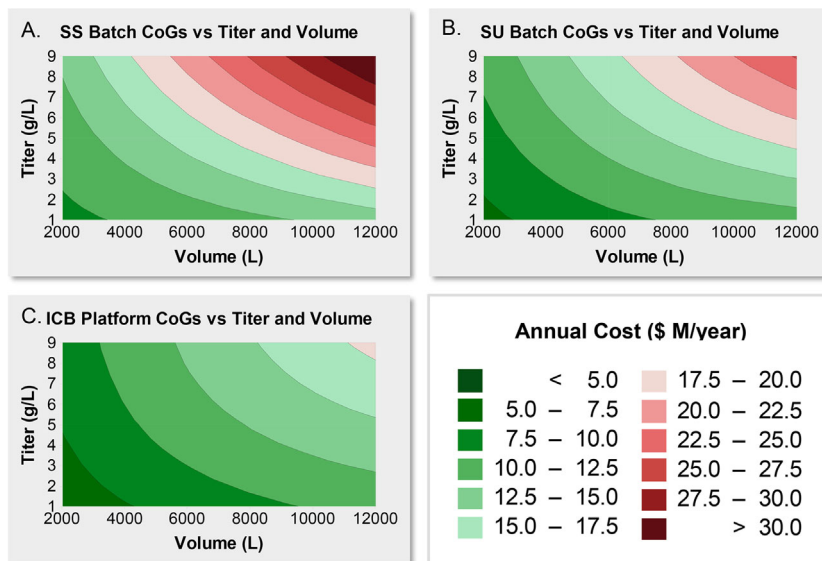


Figure 4. Contour plots of the annual CoGs associated with SS Batch (A), SU Batch (B), and ICB Platform (C) versus bioreactor titer and volume at 20 batches/year (commercial). No difference in yield was assumed between the three processes, so the annual costs are all effectively normalized by purified mAb throughput.

savings, respectively. A significant amount of these savings still come from the clarification and purification sections, as they did in the isolated clinical scenario, however the polishing section also contributes to savings here. It can be seen that the scale on the \$/g axis of Figure 5 are an order of magnitude lower than that of Figure 3. The sensitivity of the absolute \$/g values to increasing mAb throughput can be attributed largely to the fact that this is a single product analysis with particularly low facility capital utilization at the clinical scale.

The Cadence Acoustic Separator is again the source of savings in the clarification section of the ICB Platform. It offers a major reduction in capital as compared to the centrifugation used in the other two processes. Higher consumable costs for the ICB Platform clarification can be attributed mostly to a higher requirement for supporting depth filter area than what is needed after a centrifugation operation.

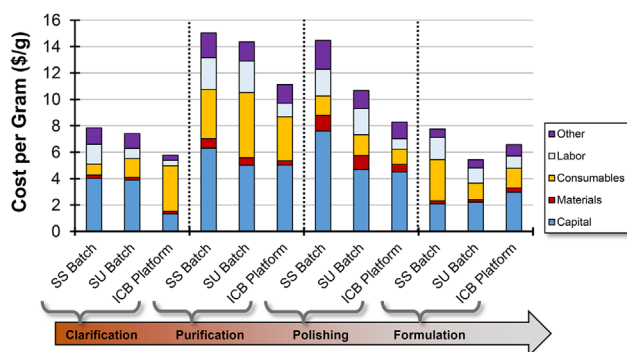


Figure 5. A comparison of the CoGs per gram of each process at a single case: 20 batches/year of 6000 L at 5 g L⁻¹. The costs are sorted by downstream section, as defined in Figure 1, and standard BioSolve CoGs category.

The Cadence BioSMB system is largely responsible for the savings seen in the purification and polishing sections of the continuous process. In both sections it contributes to labor, consumables, and materials savings over the other two processes. Consumable savings contributions are due to the BioSMB's reduction of the total required sorbent volume for bind and elute steps. Higher capacity loading, coupled with secondary load columns to avoid product loss, as well as parallel column regeneration, contribute to the BioSMB's simultaneous reduction of sorbent volume and buffer volume required per batch. Unlike the clinical scenario, savings over SU Batch due to prepacked columns are not present, because in this case buying multiple smaller columns is more expensive than a single, larger column. The Cadence BioSMB's lower buffer requirements leads to lower material costs for the buffer itself, which is most evident in the polishing section. It should be noted that for the

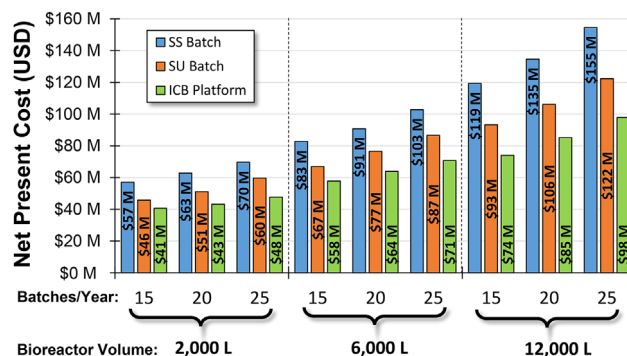


Figure 6. A comparison of the NPC for each process at a 5 g L⁻¹ titer and the full range of commercial volumes and batches/year. See section 2.5 for calculation details and approach.

flow-through polishing operations, there is no difference between the three processes in terms of the maximum cycles (60) reached by membranes, or sorbent, after 3 years. Both the SU Batch and ICB Platform process exhibit notable capital savings over SS Batch in the polishing section. This is due to the higher cost associated with buying and operating a large column as compared to that of membrane adsorbers.

The reduced labor requirements of the ICB Platform throughout all sections are due to varying combinations of shorter processing time, less buffer preparation, and fewer cleaning requirements compared to the other two processes. It is worth noting that the continuous formulation, while not contributing notably to section-specific savings, is essential for the continuous production of purified mAb in under 24 hr and the process wide savings that result in Figure 5.

The CoGs savings demonstrated by the ICB Platform across the commercial design space translate to significant total savings over the course of the project lifetime. It can be seen in **Figure 6** that even when the time value of money is considered, the total costs are the lowest for the ICB Platform across all bioreactor volumes at 5 g L^{-1} . The continuous downstream process saves \$5–25 M over the SU Batch process and \$16–57 M over the SS Batch process throughout the 14 year duration of the project. The highest relative and absolute savings for the continuous platform occur at the highest bioreactor volumes. These savings could increase if differences in the facility build time and capital spending distribution over the building period were taken into account.

4. Discussion

Batch bioprocessing using SS equipment has proven to be safe and reliable for decades and most of the recent, top-selling mAbs have been manufactured this way.^[32,41,42] SU technologies are not particularly novel, but they have typically been implemented in the small to medium scales of manufacturing for a variety reasons.^[20–24] Lately, SU technologies have become more widely used at larger manufacturing scales, with the Amgen facility in Singapore as a prime example.^[32] This should come as no surprise because we can see from multiple studies, including this one, that SU technologies can supply considerable cost savings over SS for a broad range of manufacturing scenarios.^[2,12,16–19]

The results detailed here show us that continuous processing amplifies the savings provided by switching to SU. There is a synergy between the two, because continuous processing reduces the rate at which disposable components scale with mAb throughput. While SU technologies allow for significantly reduced capital, their consumable costs can scale sharply with bioreactor titer and volume in batch operation. Combining disposable technologies with continuous processing allows for major reductions in the size of both equipment and consumables, and enables those reductions to be maintained despite increasing throughput. There are many causes for this, but in this study the simplest reason is that the same volume and product mass are spread out over a longer duration. This can be done without a significantly increased total processing duration because the continuous unit operations occur in parallel with respect to time. It is for these reasons that the ICB Platform exhibited the lowest CoGs across 78% of all scenarios modeled

here, with the greatest savings over SU Batch occurring at the highest bioreactor titers and volumes.

While the results of this study support the case for continuous bioprocessing as a way to optimize costs, they only consider a portion of the factors involved in mAb manufacturing costs. We plan to conduct additional studies to evaluate other aspects of bioprocessing, such as the impact of upstream costs, as well as the outlook for combining different upstream formats, like perfusion, with the downstream processes modeled here. Another area of focus for future upstream modeling work includes multi-bioreactor, multi-product facilities with high consumable turnover.

The process automation and in-line analytics required to run continuously for the length of a perfusion campaign will likely add costs to the continuous platform. However, process control and in-process testing would also increase responsiveness to deviations from quality targets and lead to improved consistency in product quality. This could have beneficial cost implications when process-wide risk and uncertainty are taken into account, which we hope to examine. We also plan to explore the monetary value of less obvious aspects of continuous bioprocessing, such as the potential for increased manufacturing flexibility. In addition to studies focusing on entire processes, we intend to continue work that highlights the advantages of individual technologies in response to practical challenges. An example of this would be the adaptability of the Cadence BioSMB to feed titer variability, as compared to disposable, single-column chromatography, when prepacked columns of a fixed size have already been purchased in advance.

Abbreviations

CoGs, cost of goods; ICB, integrated continuous bioprocessing; ILC, inline concentrator; ILDF, inline diafiltration; mAb, monoclonal antibody; NPC, net present cost; NPV, net present value; SS, stainless steel; SU, single-use; TFF, tangential flow filtration.

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Conflict of Interest

The authors declare no financial or commercial conflict of interest.

Keywords

continuous bioprocessing, cost modeling, downstream bioprocessing, monoclonal antibodies, single-use

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